

# **.NET Roulette: Exploiting Insecure Deserialization in Telerik UI - Caleb Gross**

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## **Content**

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# **.NET Roulette: Exploiting Insecure Deserialization in Telerik UI - Caleb Gross**

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## **Description**

DERPCON 2020 Red Team

So you're pentesting a .NET application, and you notice the server is deserializing user input—great! You know this is bad in theory, but have no idea how to actually get a shell in time for the engagement. This talk will bring you up to speed on how .NET deserialization works and how to get shells on real applications.

In this presentation, we'll dig into the internals of CVE-2019-18935, a deserialization vulnerability that allows RCE on the popular web UI suite Telerik UI for ASP.NET AJAX. After demonstrating how to exploit this issue step-by-step, you'll learn a hands-on approach to debugging a locally running ASP.NET application, quickly assessing the site's attack surface, and examining possible avenues for finding and exploiting insecure uses of

deserialization. This talk is intended for penetration testers and security researchers who'd like to begin testing deserialization vulnerabilities in .NET software.

## Transcript

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gross and today I'm gonna be talking to you about exploiting insecurity serialization and telluric UI as a bit of background about me I'm a primary US Air Force officer where I served as an offensive cyber operations lead and now I work on Bishop Fox's cast a team that stands for continuous attack surface testing where we track and test our customers external internet facing assets on a continuous basis and as a member of that team I'm focused primarily on end day vulnerability research so the CDE surfaces with no corresponding proof of exploit code I'll go and pull down the software reverse-engineer it patched it and hopefully get a working exploit that will then used to test our customers you can find me on the tweeters not breathing I'm getting this talk because I come from more of a binary network analysis background and about six months ago I began testing this bug and in a dotnet application involving insecurity serialization and I came across a lot of new concepts and terminology that I wanted to package together and deliver it for new folks who want to test this kind of bug without having to go through so much of the effort of learning like I had to do and I initially wrote a blog post but now I'm giving this talk in order to talk more about the D serialization issues and some tricks I picked up for testing asp.net applications along the way the way we're gonna break this down is first I'm going to demonstrate a vulnerability in action we're gonna get a shell on tera telluric UI for asp.net Ajax and then I'm going to break down the CDs that are at work behind the scenes in that demo we'll walk through an overview of terminology that you'll need in order to understand the nature of this vulnerability and following that we'll use what we learned to develop an exploit that will then apply to telluric UI and finally we'll walk through some things that you might come across while testing and asp.net application in real life especially in the context of testing this bug though the things we'll talk about you can extrapolate to testing p.net apps in general let's get to the demo and before I show you how the expo works I want to share a quote that I heard from have our flake recently offensive con he said exploits were the closest thing to magic spells that we experience in the real world you construct the right incantation and gay gained remote control over their bicep I thought that really captured what's been so fun for me about learning to test and security sterilization horner abilities you can get a show black magic like I'm about to show you now here what you see on the left hand side is a vulnerable instance of teller Q I'm for asp.net Ajax and that's going to represent our remote target on the right side of a couple local windows that are going to constitute our local attack server and now that I've mentioned asp.net a couple times that's a that's just a web framework that's built on the dotnet framework so telluric UI for asp.net Ajax is a popular UI suite of components for asp.net applications and it provides a lot of useful functionality including the ability to upload files asynchronously which I'll show you here so this is the normal way the software works you can say select an image and upload it to the application and attach it safe so

like a form to be sent via it you know now today instead of uploading a benign image we're instead going to upload a payload that when loaded into the application is going to give us a reverse shell it's going to call that to a server that we control and give us interactive control over the server and in order to verify that this thing is vulnerable to this exploit we're first gonna compile a payload that causes the application to sleep for 10 seconds this is going to cause a delay and the the HTTP response time which will give us an indicator that this thing is processing our code and and we can get remote code execution on it so there's the the source code compile now let's compile the payload and while it compiles in the bottom right corner I want to direct your attention to the bottom left where I've got a terminal window that's going to show us exactly when our payload gets uploaded to the server and that's just a benefit that we have just by doing this demo like this now does now that it's done compiling let's send our payload over to the server okay you can see that it uploaded successfully if we look in the bottom left we can see our DLL got uploaded to the server and right now it's being loaded into the application and we're waiting for a response and just like we expected it took about 10 seconds which is a really good indicator that this thing is going to give us a reverse shell if we upload a slightly different payload so let's instead compile a payload that instead of causing the application to sleep is going to initiate a reverse shell TCP connection back to to a server that we control and we're compiling the payload here we also need to start a netcat listener in order to receive the incoming TCP connections being initiated by the remote server so here we are starting a netcat listener and now but that's good to go let's send our payload and it's gonna happen really quick but there you can see the payload got sent over and the bottom left our DLL made it over to the remote server and we have an interactive shell and at this point we control this server according to order privilege levels that web server process was running as in this case it's running as the user tester because that's what I started the web server as and if we want to we can pop calculator we can even kill the chrome process that's running now this should demonstrate the critical nature of this vulnerability you can get remote code execution and even get a shell cool I said you a magic trick let's see what's happening in that demo behind the scenes there are two vulnerabilities at play here in telluric UI for asp.net Ajax the first one is an unrestricted file upload vulnerability that we get via a weak encryption and this is a pretty well-known vulnerability it's well documented but I still think it's important to mention it here we're gonna reference it a couple times in this talk but it's a prerequisite in order for us to then be able to exploit insecure D serialization through which we can get remote code execution and in that latter book the the insecurity realization bug is the focus of today's talk you're a visual learner I've got the exploit play that laid out here we're first going to upload a DLL a dotnet mixed-mode assembly to the server and through insecurity serialization we're gonna load that DLL into the application using telluric UI and finally that's going to give us a reverse shell I throw a couple terms at you now that you may not be familiar with like stylization and net assembly if you're wondering what all those terms mean that's okay because I was in the same boat when I started testing this these kinds of bugs a couple months ago and let's let's go ahead and start breaking down what some of those mean we're gonna take a break from talking specifically about telluric UI and and shift to talk more generally about exporting and security sterilization and the .net framework this might

be a bit of a firehose but what we're trying to do is equip you with the basic terminology and concepts that you need to exploit the insecure DC validation bug that I just showed you and while we won't talk about teller Qi just yet walk through a basic example that mimics that vulnerability made us exploited and then apply that to the real iPhone and Intel are doing now in the .net framework you have say a c-sharp code which looks a lot like Java if you're unfamiliar and that c-sharp source code is compiled to what's called the CIL code that stands for common intermediate language this is an a platform independent intermediate code that's it's between source code and machine code and it executes inside of an application virtual machine called the CLR or the common language runtime and that's a lot like Java's Cheney and if you're familiar the java virtual machine and when the CLR does is it provides portability security it prevents some common memory corruption issues like buffer overflows and and the CLR in turn takes that intermediate CIL code and compiles it at runtime via a just-in-time compiler and to native machine code that can be executed by the operating system and this is all happening in contrast to the normal C program you might be used to where you have C source that's compiled by a C compiler directly into native machine code that targets the architecture of whatever platform it's supposed to run on and and now that I've illustrated both of these things there's this concept of managed and unmanaged code that you need to understand managed code like C# that I just mentioned runs in the CLR in a software environment inside the application virtual machine and in contrast unmanaged code natively runs directly on the operating system and when I talk about a dotnet assembly this is not like x86 assembly like you might be used to hearing rather this is an exe or a DLL that contains precompiled CIL code and it also contains a manifest that details some information like the Assembly's name and version and this is the basic unit of execution for a dotnet program and and normally if a dotnet assembly only contains CIL code if it only contains managed code that runs in the CLR then that's known as a pure assembly but I want to introduce a concept called a mixed mode assembly which contains not just managed CIL code but it also contains unmanaged native code so a mixed mode dotnet assembly contains both managed and unmanaged code inside of the the same file and I want you to hang on to that term mixed mode assembly we're gonna reference it again very shortly let's shift gears for a moment and talk about serialization this is a concept you might have heard of so in in an object-oriented programming language you you may start with an object that's been instantiated and memory has stayed and has members who have values that have been manipulated and if it's out if a developer wants to take that object and maybe reuse it at a later time or send it somewhere else for use by a different application they can serialize it can burn it into a data stream that can be stored in a file in its database send over a network and sort in memory and then later on that data stream can be deserialized or converted back into an object and memory the exact structure the format of of that data stream can look pretty different depending on the programming language you're using the exact format like JSON or XML or binary but the that the developers chosen to use and also in the exact library that a developer is using - this is a bit of a it's a it was a difficult concept for me to think about abstractly and what I want to do for you is illustrate exactly what serialization looks like from the left hand side of this screen I've defined this class called a presenter and a presenter can have a first name handle a social security

number a couple attributes and then I'm then instantiating a presenter object and assigning it the first name Caleb the handle an operator and then finally I'm taking that object that I Stan she ate it and I'm serializing it first as a JSON string but then also as a byte stream because I want to show you a both of those look like here's the JSON stream you can see the B object but I instantiate it in memory and then what it looks like when it's been serialized as JSON and it looks pretty similar like you can see all of the attributes that I that I assigned to this object and now here's the byte stream and it looks a bit different what you're looking at is a is a hex junk view of the byte stream that's serialized and on the left-hand side of the raw bytes on the right hand side is an ASCII representation of that byte stream and in some ways it's similar to the to the JSON stream that I should be before but it includes this extra bit of metadata called the type of the of the object that it should be instantiated as and that informs whatever D serialization program that's converting this data stream back into an object exactly what type of object it should be instantiated as and now I've written a program that's going to simply take a a serialized JSON stream from the command line and deserialize it and reinstate a presenter object so we first CC realized that stream and and then cast it as a presenter object because it's the type of object that we're expecting to receive and and so now I'm passing two different json streams to this deserialize ER program that i just wrote the first one is is the same json stream that i had showed you two slides ago but I'm adding a bit of extra data here I added a type fueled that tells the big deserializing program oh that this is a presenter type and and so when IDC realized that that stream midas realizing program takes that object and prints out the the presenters handle just to verify that we successfully deserialize this thing so everything everything looks good right here next if you look further down on the screen I'm passing another JSON strange the same program but this time I'm not passing a presenter type I'm instead passing an assembly Installer site and instead of passing attributes like first name and handle I'm passing a single member called a path and and that's a path to to a dll and we would expect that because this is expecting our DCO lser program is expecting a presenter type but it should just complain and say hey you passed me an assembly installer when I'm at when I'm expecting it presenter and so I'm not gonna deserialize this but that's not what happens instead you see an exception here that says hey I tried to load an assembly a.net assembly a dll and I couldn't find it and that's because it didn't exist but I wanted to point out that that this error tells us that it's actually trying to instantiate this assembly installer object and loaded diello into memory which is super interesting and let's see if if we can find a way to exploit this there's a bit of prior research but I think you ought to be aware of if you're gonna start looking at and secured deserialization vulnerabilities a few presentations I want to point out our first marshaling pickles this is a presentation that happened in 2015 and it got the ball rolling on lot of conversation around attacking insecurity sterilization and some of the response to this talk is like okay let's but stop using Java D sterilization and instead try to use supposedly secure json libraries which at the time were free of known RCE vectors and a couple years later this presentation came around called Friday the 13th JSON attacks where these researchers showed that these JSON libraries were vulnerable as well and in the process they identified an attack vector in dotnet for exploiting and security serialization that we're gonna use and exploit today and then finally as this relates to tell rqi specifically there are

a couple people who did some work that was pretty instrumental and setting the air up for success as I investigated this T serialization bug first Paul Taylor wrote an exploit for the prerequisite unrestricted file a cloner ability and also developed a custom payload feature for the D serialization vul that Marcus will change reported about a year ago and and this is what the scene looked like when I started investigating this vulnerability there was Marcus's third-party security advisory for insecurity sterilization and telluric UI and he talked a bit about it but there wasn't any POC exploit code that should exactly how to get arbitrary code execution and in my role doing n day research for Bishop boxes caste service I took a closer look and was able to work out a proof of concept exploit to get remote code execution like I'm going to show you let's continue where we left off NAR preached honor in our previous example we were able to specify an arbitrary type and JSON object the program wasn't expecting and when the when we specified that our program should deserialize an instant she and assembly installer and provided a path to a dll that resulting object try to load that DLL into memory and the fact that it behaves that way means that assembly and the class assembly installer is what we call addy serialization gadget that is it's a class within the applications executing scope that when is when it's deserialized and instantiated it has some special side-effects that make it useful for an attacker and in particular assembly assembly installer is what we'll call an RC e gadget because it performs some operations when setting its path member that that facilitate executing arbitrary code so now that we understand that we can load a dotnet assembly into the application by specifying the path attribute and ng sterilizing an assembly installer streamlets what's soon as I can do for us as documented and the Friday the 13th JSON attacks presentation we can create and craft a mixed mode assembly to remember that term it contains managed and unmanaged code and inside of that assembly that DLL we can put arbitrary code inside the DLL main function that's the dll's entry point and it's going to be triggered when the assembly is loaded into memory in note that it's located in the unmanaged section of the mixed mode assembly inside the native code section and typically when you if you're if you want to create a mixed mode assembly you can write that in C++ and it's pretty straightforward but we're not going to do that we're instead going to use C in order to write our unmanaged native portion of the mixed mode assembly and that's because I've encountered situations it's pretty rare but there are targets but I've tested that aren't able to execute code that was compiled from C++ and and that's likely because it didn't have a C++ this redistributable installed meaning it just doesn't have the software it needs to execute a code that was compiled from C++ so instead we're going to use C and and that means it's a bit less straightforward we'll have to compile our mixed mode assembly in sort of a piecewise fashion first by taking some some dummy like standin managed code I'm sharing here on the right hand side it's just a simple empty class it doesn't do anything but it's important that we that we use it because for for our mixed mood assembly to be recognized as a valid medicine we it needs to contain both managed and unmanaged code so we had this empty class to constitute the manage section and then we compile a malicious payload and see where we put our code inside the DLL main function and then link both of those together to form our mixed-mode assembly I've shown you an overview of how that compilation process works but I wrote a script that does all that for you and takes care of of creating your assembly I just wanted

to show it to you here but I've included a link to github and you can look at the full script there now let's see how we can get remote code execution by crafting and loading a DLL dammit assembly into its memory well first write our payload here on the left hand side and inside the D element DLL main function you can see I'm just including a message box that's going to save poem when this thing is executed then I pass that source code to my build script that I just mentioned which generates a dotnet mixed-mode assembly DLL and then finally I'm taking the JSON stream and specifying that our target program D serializer should should deserialize and instantiate an assembly installer object whose path is pointing to message box DLL the payload that we just generated and when we run a program we see our message box pops just fine even though you see in the background that the the program is still throwing an exception and saying hey you passed me an assembly installer but I was expecting to cast that as a presenter even though we get an exception our code still works this is a really important point to note here this is because the process of this stream being deserialized and the object being instantiated although that happens before that resulting object is attempted to be cast as a presenter which means we still get remote code execution and it's really cool you should also note the main issue here and what makes this application vulnerable is that the program is deserializing inputs untrusted and put directly from a user and here I'm passing that on the command line and in a real vulnerability that you might find at an actual application it's likely to be passed in a different way like perhaps over the network so I just wanted to point that out here as well ok that was a lot we talked about DC realization we defined what a dotnet mixed-mode assembly is that it contains managed and unmanaged code and we shadow we can craft a payload and unload it into a target application via an RC e gadget that will facilitate execute on an arbitrary code as long as we control the type and the object that's being deserialized and now equipped with the understanding of those terms and concepts that I introduced and the example exploit that I showed you let's go exploit a real-life vulnerability and Teller Qi in the beginning of the presentation I showed you a demo where I first uploaded a file and when when TELRIC you I uploads file it accompanies the contents of that file with a the serialized configuration object an async upload configuration and in that configuration specifies things like where the file should be uploaded to the illogic file extensions that you're allowed to upload and and it takes that object the serialized object and it encrypts both the JSON stream and the type both of which I've explained to show you in an example before and and it sticks them together with an ampersand in the middle and sends it over to the server I'm showing you here the decrypted version of both the JSON object and the type string just so you can see what those look like and to recognize that it looks similar to the present example that I showed you before here's the bit of server-side code that's receiving that that encrypted serialized async upload configuration and what this does is it first splits it apart based on the ampersand character that I mentioned and decrypts the object and type and then passes them both to a deserialized function a DC realizes and instantiates the async upload configuration object and then then finally casts it you can see on the bottom left as an async upload configuration because that's what it's expecting and here's the actual function that's doing the deserializing it looks a bit different from from the very simple example that i showed you but the core thing going on here is the same that this javascript serializer is invoking

it's deserialized method on a on a type and an object and we've been able to specify specify both of those values by by crafting a file uploaded request to tell Rick UI let's see if we can get remote code execution just like we did in the presenter example I'm repeating and revisiting the slide here that I should you near the beginning of this presentation just remind you of the exploit flow we're first uploading a payload or dotnet mixed-mode assembly DLL and then loading it into the application by using an assembly installer type and while deserializing and then finally getting an interactive shell and and you should recognize we're triggering the same logic twice that server-side code I just showed you we're first passing a normal file upload configuration like it expects to accompany our RDL that we're uploading and then finally were we're instead passing it instead of an async upload configuration realized object we're passing a serialized assembly installer to use as an RCD gadget just like we saw in the demo we first caused this application to sleep our C payload is going to contain a yellow main function with not messagebox but sleep and in that function and and that's gonna cause it delay in the HTTP response it's really useful to verify this vulnerability because there might be strong egress filtering on the network and this may be the only way that you can see that it's blindly executing our code and once we compile it with the build script but I showed you as well we're gonna send it over and deliver it to the target application yeah a Python script but I also wrote and put on github I'm not going to show the full contents of that script here it handles a lot of encryption and file upload nuances as well as formatting the D serialization logic and so that the target application will process it correctly you can see here that on on the bottom of the screen the bottom right we get a delay of 10 seconds which indicates that this thing is processing our our arbitrary code and and now instead of causing it to pop a message box or or sleep we're gonna craft a different payload but it'll cause this thing to spawn of thread and launched an interactive shell and connect us to it by via a a reverse TCP connection I have sort of an abbreviated payload here but I just wanted to highlight exactly how it's working and and then once we upload that payload and load it into the application we'll catch the the reverse TCP connection with a net cap that we have listening on them on our attack sir and here's how it works you saw in video form but but I'll repeat here one more time we use that Python script after compiling our payload and we upload the DLL with an initial request and then following up with a second request we specify an assembly installer object that the program should be serialize which whose path points to the dll we just uploaded and uploads it into memory and calls back to us with perverse interactive show and and again I want to highlight this is all happening because we can specify an arbitrary JSON object and type to the deserialized function that's taking place on on server side this is fixed in later versions of TELRIC UI now they use a whitelist to restrict the classes or types that can be deserialized with that json stream so it'll allow you to deserialize and instantiate an async upload configuration object but it's not gonna let you specify assembly installer to be serialized and that's trying to be pretty pretty effective to mitigate this vulnerability while developing this exploit a mitigating a few possibilities for her exploit to deny service to the to the target web application the first way I've already showed you I threaded the DLL payload it launches its reverse shell logic as a thread and the reason for doing that is that it avoids blocking the web apps user interface while the reverse shell process is running because otherwise the application would just kind of hang

and wait for our shell to exit before returning a response and and and it would block legitimate users from from using the web application in the noon time but now since that kind of backgrounds are reverse shell we can use the application and so can the legitimate user and and then I also rename the uploaded assembly on disk remotely and and the reason for doing that is because telluric UI the way it uploads files say you upload two files of the same name one file doesn't overwrite the other but instead the second file is appended to the first one that got uploaded and that can form what ends up being a malformed medicine we think you crash the web server when it's attempted to be loaded into memory so you should always consider when developing and exploit what can go wrong here because you really don't want to cause a denial of service at this point we know how to exploit the vulnerability I'm gonna show you what it looks like to to exploit it while dealing with all the new ones that a real target provides we're gonna walk through some tips and tricks that I picked up for exporting asp.net web applications and the context of testing this and security serialization bug and the theme for a lot of the things I want to talk about is that you shouldn't throw an exploit expecting it to just work we're taking functionality say that's intended for a file upload and we're getting remote code execution that's that's a really tricky thing and you should be prepared to tailor the attack to the specific target that you're testing and that's re it can vary in terms of the specific software version that's running the dotnet framework version the underlying architecture egress filtering rules on the network and other configuration differences that cause it to behave in a way that you might not expect it very first thing that I would always recommend doing when testing any sort of application to include an asp.net web app is to download a local copy of the software and and you should try to look for the exact version exact architecture everything as compared to the target that you're testing and once you've pulled it down you can run it what I like to do is use a is Express this is sort of a lightweight version of iOS that allows you to quickly get an asp.net web app up and running without filling with a lot of the Machine wide settings that you often have to deal with and in is and then once you've got this thing running on your local server you can debug it with an open source application called DN spy this is a dotnet d compiler and debugger and my tool of choice for this task you can attach it to your running web application and do things like examine the modules or dotnet assemblies that it's loaded into memory and and you can see one of those that I've highlighted here teller that web UI that web UI dot DLL that's where our vulnerable code resides you can also search for for useful entry points into the application here I'm searching for the function process request and it's one that I often look for because it's sometimes the first place that a developer is going to start writing custom code to receive and process an HTTP request that you sent to the application and and therefore it's going to be one of the first places that might receive user input that that you control you can also take I took one of the process request functions and open it up in a decompile view in in this case what you're looking at is a third-party component called client dependency that's in charge of fetching like static resources for a web application and it has a an I a-- v vulnerability that lets you read an asp.net applications web config file and normally what I do here is enter the process request function and add a breakpoint right at the beginning of that function so that when I start testing this this component that the debugger will stop and pause execution as soon

as it hits that function and I can then inspect the values of what the local variables that are that have values assigned to them and see exactly what's going on while testing code like this I mentioned the web config just a moment ago this is a file that every asp.net application has and you should always check to see if it's exposed to the Internet it shouldn't be because it contains sensitive configuration information when disclosed can allow you to do things like attack you state the serialization that's sort of out of the scope of this talk but it's important to mention here because it's good it's pretty relevant but what that can take also contains a few HTTP handlers that I've showed you here on the right hand side and these handlers map directly to the process request function as I showed you a moment ago you can see there's a process request function for the async upload handler where where our vulnerable code resides and you can see the same thing highlighted down here this telluric that web UI that a resource path is mapped to an assembly that contains that process request function so this is a good place to look because these HTTP handlers often exposed third-party components that bypassed the core asp.net web page model they're in charge of like loading static resources and doing other things kind of dynamically in a lightweight way and therefore might not get as thoroughly tested or looked at by developers or penetration testers and is a good place to look for horrible code uh I'd also encourage you always to pay attention to robust error messages like exception details and stack traces because this can do closed disclose like the paps the web route which is really good to know if you want to upload a web shell a stack trace can also show you exactly where your attack is is breaking and I've use that many times in order to say like realize that the target web server is using an older version of the .net framework than then the version that I used to compile my payload which made it incompatible and and so I had to adjust and treat my attack according to the target that I was testing another really cool thing you can do is pay attention to if you if you need to signature the exact version of the software using you should look at a last modified a and e tags because those can indicate when the software was released if you're unfamiliar with an e-tag sometimes it's it's structured in the format I noted size timestamp so in this case you have a timestamp there on the right which you can convert from hex to decimal and at the time and then convert that epoch time to a date string that you can read and now I'm showing you an example here intel rqi the version string is disclosed and a resource that we're requesting and so that's really useful but if you don't find a version string there Salem cut is a last modified day and in this case it's May 3rd 2017 as you can see that let's hang on to that and then take a look at torus release history for the software you can see on the exact same date May 3rd 2017 that released through 2017 that 2.5 to 3 which tells us that this target is very likely running that version the reason that happens is because when software vendors release software they archive it which preserves the last modified date on disk at the time that they are content and then when you later on install and extract that software your machine that modified that modification date is preserved when the software is installed and then returned to you in the form of a last modified date when you request like a static resource but say that we don't have a version string or or last modified date you can sometimes brute force dotnet assembly versions like we need to do for the prerequisite under circuit file upload the owner ability and while this can normally be kind of a laborious process and you have to iterate through like hundreds of different

versions I realize about testing the software that you can sometimes get away with just specifying the major version of the assembly in this case that's just 2017 and and you can see that as I'm uploading my payload I'm only providing the version 2017 and the server responds up yeah cool I'm running version in 2017 back to that 503 at 40 and I want of as long as there aren't any types that are conflicting with it that I worked just fine so keep that in your in your tool belt as well say say you can't upload your payload or maybe you can upload it but you but you can't load it into memory for some reason as a last resort what you can do is set up a responder server which starts a server that can receive incoming SMB connections and an issue an attack on D serialization like I've showed you with assembly installer but instead of specifying that its path should point to a local DLL that's a that's been uploaded to the remote server you can point that path via a UNC path back to the responder server that you set up and even the dotnet framework has some restrictions around loading assemblies from remote locations it often won't let you do it it'll still go out and try to fetch it at least even though it won't load it into memory and what that means as it reaches out with an SMB connection and that allows responder to dump ntlm hashes for the server that you're attacking and I've used that a couple times when I wasn't able to actually get remote coded code execution on the host only share a story which is also the namesake of this presentation aye aye one point I was testing an instance a teller Qi and I uploaded a mixed mode assembly but strangely I couldn't loaded into memory I couldn't really hear what was going on so I watched the error messages that were coming back and I saw that it was disclosing the internal host name of the of the host that I was attacking and that the host name when I uploaded a payload was different than the host name that I was seeing when I tried to load it into the application via insecurity serialization and after a couple minutes I realized I'm attacking three load balance servers and I had to repeatedly attempt a DC realization attack over and over and kind of a round robin fashion until I finally hit the same post that I uploaded my halo to and and that's what brought me over the finish line by paying attention to verbose error messages got just a couple takeaways for you peace realization bugs they continue to emerge and I think a lot of developers and pen testers still don't fully understand them I hope that my presentation today has demonstrated but this is really worth understanding at a fundamental level and applying a thorough approach when you're testing applications that deserialize untrusted data specified by a user but the risk of saying it too much always watch error messages it's I can recall so many times when that's finally allowed me to exploit an application that I was really having a difficult time with and then finally these serialization it's pretty one word it's a bit difficult to say and even harder to write it over and over again and in a presentation like this I think it's a great candidate to be abbreviated with the numeron in d13 n just like we abbreviate kubernetes with Kas and I just tend to think about that's all I've got thank you so much for your listening you can find links to my exploit that I wrote on github and then also the blog that I wrote about this both a couple months ago have any questions please leave them in the right team talks disk or channel and have a great day thanks a lot